

Muon-Induced Background in the Unanalyzed Years of the DM-Ice17 Dark Matter Detector

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1. Background and State of the Field

Dark matter makes up roughly a quarter of the universe’s energy content, yet it has never been detected in a laboratory. For more than two decades one experiment has claimed otherwise. DAMA/LIBRA, operating beneath the Gran Sasso mountain in Italy, uses crystals of thallium-doped sodium iodide, NaI(Tl), a material that emits a brief flash of light when a particle deposits energy in it. DAMA reports that its rate of very low-energy flashes rises and falls once per year, peaking near June 2, and interprets this as the Earth’s orbital motion carrying its detector through a galactic halo of dark matter particles: in June the Earth moves into the oncoming wind of dark matter, in December it moves with it, and the collision rate should follow [1].

No other experiment has confirmed the result, and the central objection is mundane. Laboratories also have seasons. Temperature, humidity, radon concentration, and the flux of cosmic-ray muons through the rock all vary annually, and any of them could imprint a yearly cycle on a detector. Distinguishing a galactic signal from a terrestrial one therefore comes down to *phase*. A dark matter signal is set by the Earth’s orbit and keeps its June maximum everywhere on the planet, while most environmental effects follow the local seasons and invert in the southern hemisphere.

DM-Ice17 was built to exploit exactly that asymmetry [2]. It consists of two 8.5 kg NaI(Tl) crystals deployed in 2010 at the bottom of two strings of the IceCube Neutrino Observatory and frozen permanently into the Antarctic ice at 2450 m depth. IceCube instruments a cubic kilometre of that ice with thousands of light sensors in order to detect neutrinos [3]; muons—heavy, short-lived relatives of the electron, produced when cosmic rays strike the upper atmosphere—penetrate deeply enough to reach the array constantly and are IceCube’s dominant background. For DM-Ice17 this arrangement is a gift: it is the only NaI dark matter detector ever operated inside an active cubic-kilometre muon telescope, so the particles passing through its crystals can be identified independently by the surrounding array.

The published DM-Ice muon analysis established three things [4]. Muons cross each crystal about 2.93 times per day; that rate modulates by $12.3 \pm 1.7\%$ and peaks around January 22, in austral summer, four months from the DAMA phase; and muons leave behind a faint afterglow in the crystal, phosphorescence with a decay time of 5.5 ± 0.5 s, which the authors characterized as noise and argued cannot reproduce the DAMA signal. Crucially, that analysis used 3.5 years of data and stops in January 2015, two months after the crystals' photomultiplier gain was raised roughly tenfold. Twelve years of data were taken. Fewer than four have ever been analyzed, and the unanalyzed years are the ones the modern dark matter search would use.

2. Objective, Significance, and Implications

My objective is to extend the characterization of muons and muon-induced background into the unanalyzed 2015–2020 high-gain era, and to convert the known existence of afterglow into a quantitative statement about a dark matter search: **what fraction of events in the 2–6 keV electron-equivalent (keVee) signal region is muon-induced, what fraction of that an IceCube veto removes, and at what cost in lost exposure.**

This goes beyond the published work in two specific ways. First, coverage: the tenfold gain change did not merely shift a calibration, it inverted the behavior of the pulse-shape parameter used to identify muons, which is why no selection was ever published for those years. I have already solved that problem, so the era is now open for the first time. Second, framing: the published measurement established that phosphorescence exists and decays in about 5.5 seconds, but treated it as a noise source rather than asking how much of it survives a low-energy event selection and lands inside the signal region. Those are different questions, and only the second one bears on whether a June-phase signal could be environmental.

I want to be direct about what this cannot do. With 17 kg of crystal, DM-Ice17 will never outmeasure DAMA/LIBRA's roughly 250 kg at a lower threshold, and this project will not produce a competitive dark matter limit. Its value is that DM-Ice17 is the only place where the muon-induced contribution to a NaI signal region, and the effectiveness of an active veto against it, can be *measured* rather than simulated—numbers that feed directly into the successor experiment, COSINE-100 [5],

and into any future veto-instrumented NaI deployment.

3. Research Plan and Methodology

Preparation already completed. This is a continuation of work I began last summer, and its infrastructure is built and validated. I reduced the full 2011–2022 dataset—2.38 billion recorded events—into a compact columnar form, and built the first true live-time database for DM-Ice in the group’s code base, covering 95,174 runs. Applying the published method to the low-gain era, I recover a muon rate of 2.918 ± 0.038 per crystal per day and a flux modulation of $10.2 \pm 1.6\%$ peaking on day 30 ± 9 , all consistent within uncertainties with the published 2.93 ± 0.04 , $12.3 \pm 1.7\%$, and January 22 [4]—my evidence that the pipeline is trustworthy. I then diagnosed why a naive transfer of that selection to the high-gain era overcounted muons by a factor of 751, and fixed it with a discriminant not previously used on these data, reaching 0.95 efficiency and 0.93 purity.

Step 1 (weeks 1–2): establish the low-energy threshold. My own calibration notes flag that a feature previously identified as a 46.5 keV lead-210 line is more likely the detector’s trigger turn-on, its width being far too narrow for a real line. Because everything downstream depends on it, I will first measure where the turn-on actually lies and define the lowest defensible analysis band from the data rather than from the literature.

Step 2 (weeks 3–6): measure afterglow leakage into the signal region. For each of roughly a thousand tagged muons per year per detector, I will count low-energy events in delayed windows following the muon and compare them against randomized control windows drawn from the same runs. Because both window classes are drawn from identical live-time, the comparison is self-normalizing and immune to the live-time uncertainties that dominate any rate measurement. This yields the muon-induced fraction of the signal-region rate and its decay profile, which I can check against the published 5.5 s constant as a validation.

Step 3 (weeks 7–9): veto efficiency and its cost. Using the muon sample tagged by IceCube, I will compute what fraction of that leakage an active veto removes as a function of the veto window length, together with the exposure sacrificed. At three muons per crystal per day, even a 30-second window costs roughly a tenth of a percent of exposure, so I expect the interesting result to be how

much background it buys.

Step 4 (weeks 10–13): extend to the high-gain era. Low-energy events in the high-gain years are digitized on a channel my current reduction does not carry, so extending Steps 2 and 3 past 2015 requires a re-extraction that I have written and tested on a subsample. This is the novel-coverage half of the project and I will launch it in week 1 to run unattended, but I have deliberately placed it after a complete low-gain result so that a storage bottleneck cannot cost me the term.

Step 5 (weeks 14–15): one open systematic, and write-up. The 2015 muon rate sits well below 2016–2020 and is not a selection artifact; I will test whether per-run live-time is overestimated because it cannot resolve dead time within a run. I will then write an internal collaboration note and draft a conference abstract.

4. Timeline

The Harvard term runs from early September to mid-December, and I can commit 10 to 12 hours per week. Long extractions run unattended on IceCube’s Cobalt and NPX computing systems, so the heaviest computation does not compete with coursework. The five steps above are scheduled across weeks 1–2, 3–6, 7–9, 10–13, and 14–15 respectively, with two review points: a measured threshold at the end of week 2, and the muon-induced leakage fraction at the end of week 6, each discussed with my mentor before I build on it. I will present Steps 2 and 3 to the group in early November. The project is deliberately ordered so that a complete, self-contained result exists by week 9 even if the high-gain extension in Step 4 does not finish.

5. Faculty Involvement and Support Network

My faculty mentor is Dr. Carlos Argüelles-Delgado, who leads Harvard’s IceCube group and has supervised this project throughout. I attend his weekly group meetings on Tuesdays from 3–5pm, where the state of the whole lab is discussed, and I will meet with him one-on-one at least four times over the term, including at the two review points above. He sets the physics priorities, and he has previously required me to freeze and defend a completed analysis in writing before moving on—a practice that produced the memo this proposal is built from. He also provides the access the

project depends on: IceCube’s Cobalt and NPX systems, the group’s allocation on the FAS research cluster, and the IceCube Collaboration channel through which any result of mine receives internal review before publication.

For day-to-day technical guidance I have weekly check-ins, at minimum, with Dr. Felix Yu, a postdoc in the group whose work on reconstruction methods overlaps my own and who defined several of the tasks I completed last summer; I will adjust that frequency to my needs as the term progresses. Working in the department’s Laboratory for Particle Physics and Cosmology also puts me among the other students on the team, which has been the most reliable way I have found to learn techniques that are not written down anywhere. One question I will bring to my mentor early is whether to frame the term’s output primarily as a background-characterization result for the collaboration or as a modulation measurement, since that affects how much weight Step 4 should carry.

6. Citations

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